HIGHER CHERN MONOMIALS VANISH FOR CLOSED STRICTLY PSEUDOCONVEX CR MANIFOLDS

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Allowed standard prerequisites (editorial).

- Research prerequisites named in the original article. Full original bibliography retained; all source-local definitions and supporting arguments retained.

1. Introduction

Let $\big(M, T^{1,0}M\big)$ be a closed connected strictly pseudoconvex CR manifold of dimension $2n+1 \geq 5$. It is known that M can be realized as the boundary of a strictly pseudoconvex domain in a complex manifold [1, 6, 8]. This fact gives some restrictions to its topology. For example, Bungart [2] has proved that the cup product

$$H^{i_1}(M, \mathbb{C}) \otimes \cdots \otimes H^{i_m}(M, \mathbb{C}) \to H^{|I|}(M, \mathbb{C})$$

vanishes for any multi-index $I = (i_1, \ldots, i_m) \in \mathbb{N}_+^m$ satisfying $i_l \leq n-1$ and $|I| = i_1 + \cdots + i_m \geq n+2$; see also [12]. As noted in the last paragraph of [2], this result follows also from a result of L^2 Hodge theory on strictly pseudoconvex domains by Ohsawa [10, 11].

In this paper, we will point out that Ohsawa's result gives also a constraint on Chern classes of strictly pseudoconvex CR manifolds. For a complex vector bundle E and a multi-index $K = (k_1, \ldots, k_m) \in \mathbb{N}_+^m$, we denote by $c_K(E)$ the cohomology class $c_{k_1}(E) \cdots c_{k_m}(E)$.

Theorem 1.1. *Let $\big(M, T^{1,0}M\big)$ be a closed strictly pseudoconvex CR manifold of dimension $2n+1 \geq 5$. Then $c_K\big(T^{1,0}M\big)$ vanishes in $H^{2|K|}(M, \mathbb{C})$ for any multi-index K with $2|K| \geq n+2$.*

Note that sharper results than Theorem 1.1 hold for some particular cases. For instance, Theorem 1.1 holds for $\mathbb{Z}$-coefficients if M can be realized as a real hypersurface in a Stein manifold. On the other hand, we can relax the degree condition to $2|K| \geq n+1$ if M is the link of an isolated singularity; see [7] for example. However, we will see in Section 4 that Theorem 1.1 is "optimal" for a general closed strictly pseudoconvex CR manifold. More precisely, we will give an example of a closed strictly pseudoconvex CR manifold $\big(M, T^{1,0}M\big)$ of dimension $2n+1$ with $c_K\big(T^{1,0}M\big) \neq 0$ in $H^{2|K|}(M, \mathbb{Z})$ for any multi-index K with $1 \leq |K| \leq n$ (Proposition 4.1). We will also construct a $(2n+1)$-dimensional closed strictly pseudoconvex CR manifold $\big(M, T^{1,0}M\big)$ with $c_K\big(T^{1,0}M\big) \neq 0$ in $H^{2|K|}(M, \mathbb{C})$ for any multi-index K with $1 \leq 2|K| \leq n+1$ (Proposition 4.2).

Theorem 1.1 in particular implies that $c_2\big(T^{1,0}M\big) = 0$ in $H^4(M, \mathbb{C})$ for any closed strictly pseudoconvex CR manifold $\big(M, T^{1,0}M\big)$ of dimension five. Hence the assumption on the second Chern class in [4, Proposition 8.8] automatically holds for the strictly pseudoconvex case, which is compatible with [9, Theorem 4.6]. It would

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be an interesting problem whether Theorem 1.1 remains true for non-degenerate CR manifolds or non-compact strictly pseudoconvex CR manifolds.

We also remark a relation between Theorem 1.1 and contact topology. Let M be a closed oriented $(2n+1)$-dimensional manifold with a positive co-oriented contact structure ξ. Then we can define the k-th Chern class $c_k(\xi) \in H^{2k}(M,\mathbb{Z})$ of ξ by using an adapted almost complex structure on ξ. A contact structure ξ is said to be *holomorphically fillable* if M can be realized as the boundary of a strictly pseudoconvex domain and $\xi = \operatorname{Re} T^{1,0}M$ holds. In this case, $c_k(\xi)$ coincides with $c_k\big(T^{1,0}M\big)$. Therefore Theorem 1.1 implies the following

Corollary 1.2. *Let (M,ξ) be a holomorphically fillable contact manifold of dimension $2n+1 \geq 5$. Then $c_K(\xi)$ is a torsion class in $H^{2|K|}(M,\mathbb{Z})$ for any multi-index K satisfying $2|K| \geq n+2$.*

This result is no longer true for a general contact structure. As far as the author knows, such a constraint on Chern classes has not been obtained yet for a contact structure satisfying other fillability conditions.

This paper is organized as follows. In Section 2, we recall some basic notions related to strictly pseudoconvex domains and CR manifolds. Section 3 provides a proof of Theorem 1.1. In Section 4, we will see that, by constructing some examples, Theorem 1.1 does not hold for $\mathbb{Z}$-coefficients in general, and the degree condition is optimal.

2. Preliminaries

Let X be a complex manifold of complex dimension $n+1$. A real-valued smooth function φ on X is called an *exhaustion function* if $\varphi^{-1}((-\infty,c])$ is compact in X for any $c \in \mathbb{R}$. We say φ to be *strictly plurisubharmonic* if its complex Hessian $\big(\partial^2\varphi/\partial z^i \partial \overline{z}^j\big)$ is a positive-definite Hermitian matrix for any holomorphic local coordinate $\big(z^1,\dots,z^{n+1}\big)$. A *Stein manifold* is a complex manifold admitting a strictly plurisubharmonic exhaustion function.

Let Ω be a relatively compact domain in X with smooth boundary $M = \partial\Omega$. Then there exists a smooth function ρ on X such that

$$\Omega = \rho^{-1}((-\infty,0)), \qquad M = \rho^{-1}(0), \qquad \mathrm{d}\rho \neq 0 \text{ on } M;$$

such a ρ is called a *defining function* of Ω. A domain Ω is said to be *strictly pseudoconvex* if we can take a defining function ρ of Ω that is strictly plurisubharmonic near M. We call Ω a *Stein domain* if Ω admits a defining function ρ that is strictly plurisubharmonic on a neighborhood of the closure of Ω. Note that a Stein domain is a Stein manifold; this is because $-1/\rho$ defines a strictly plurisubharmonic exhaustion function on Ω.

We next recall some notions related to CR manifolds. Let M be a $(2n+1)$-dimensional manifold without boundary. A *CR structure* is an n-dimensional complex subbundle $T^{1,0}M$ of the complexified tangent bundle $TM \otimes \mathbb{C}$ satisfying the following two conditions:

$$T^{1,0}M \cap \overline{T^{1,0}M} = 0, \qquad \big[\Gamma\big(T^{1,0}M\big), \Gamma\big(T^{1,0}M\big)\big] \subset \Gamma\big(T^{1,0}M\big).$$

A typical example of a CR manifold is a real hypersurface M in a complex manifold X; it has the natural CR structure

$$T^{1,0}M = T^{1,0}X|_M \cap (TM \otimes \mathbb{C}).$$

A CR structure $T^{1,0}M$ is said to be *strictly pseudoconvex* if there exists a nowhere-vanishing real one-form θ on M such that θ annihilates $T^{1,0}M$ and

$$-\sqrt{-1}\mathrm{d}\theta\big(Z,\overline{Z}\big) > 0, \qquad 0 \neq Z \in T^{1,0}M;$$

we call such a one-form a *contact form*. The natural CR structure on the boundary of a strictly pseudoconvex domain is strictly pseudoconvex. Conversely, any closed connected strictly pseudoconvex CR manifold of dimension at least five can be realized as the boundary of a strictly pseudoconvex domain.

3. Proof of Theorem 1.1

Let $\big(M, T^{1,0}M\big)$ be a closed strictly pseudoconvex CR manifold of dimension $2n+1 \geq 5$. Without loss of generality, we may assume that M is connected. Then it can be realized as the boundary of a strictly pseudoconvex domain Ω in an $(n+1)$-dimensional complex projective manifold X [8, Theorem 8.1]. The vector bundle $T^{1,0}X|_M$ is decomposed into the direct sum of $T^{1,0}M$ and a trivial complex line bundle, and consequently $c_K\big(T^{1,0}M\big) = c_K\big(T^{1,0}X|_M\big)$ is in the image of the restriction morphism $H^{2|K|}(\Omega, \mathbb{C}) \to H^{2|K|}(M, \mathbb{C})$. On the other hand, the natural morphism $H^i_c(\Omega, \mathbb{C}) \to H^i(\Omega, \mathbb{C})$ is surjective for $i \geq n+2$ according to [11, Corollary 4]; see [2, Lemma] for another proof. From the exact sequence

$$H^i_c(\Omega, \mathbb{C}) \to H^i(\Omega, \mathbb{C}) \to H^i(M, \mathbb{C}),$$

it follows that $H^i(\Omega, \mathbb{C}) \to H^i(M, \mathbb{C})$ is identically zero for $i \geq n+2$. This completes the proof of Theorem 1.1.

4. Examples

In this section, we treat some examples related to Theorem 1.1. We first show that Theorem 1.1 does not hold for $\mathbb{Z}$-coefficients in general.

Proposition 4.1. *For each positive integer n, there exists a closed strictly pseudoconvex CR manifold $\big(M, T^{1,0}M\big)$ of dimension $2n+1$ such that, for every multi-index K with $1 \leq |K| \leq n$, the K-th Chern class $c_K\big(T^{1,0}M\big)$ is not equal to zero in $H^{2|K|}(M, \mathbb{Z})$.*

Proof. Let $\mathbb{C}P^n$ be the n-dimensional complex projective space and $\mathcal{O}(-d)$ be the holomorphic line bundle over $\mathbb{C}P^n$ of degree $-d < 0$. Fix a Hermitian metric h on $\mathcal{O}(-d)$ with negative curvature, and consider

$$M = \big\{v \in \mathcal{O}(-d) \,|\, h(v, v) = 1\big\},$$

which is a principal S^1-bundle over $\mathbb{C}P^n$. This manifold has a strictly pseudoconvex CR structure $T^{1,0}M$ induced from the total space of $\mathcal{O}(-d)$, and $T^{1,0}M$ is isomorphic to $\pi^*T^{1,0}\mathbb{C}P^n$ as a complex vector bundle, where $\pi\colon M \to \mathbb{C}P^n$ is the canonical projection. Consider the Gysin exact sequence

$$H^{i-2}\big(\mathbb{C}P^n, \mathbb{Z}\big) \xrightarrow{-d\cdot\alpha} H^i\big(\mathbb{C}P^n, \mathbb{Z}\big) \xrightarrow{\pi^*} H^i(M, \mathbb{Z}) \to H^{i-1}\big(\mathbb{C}P^n, \mathbb{Z}\big),$$

where $\alpha = c_1(\mathcal{O}(1))$ is a generator of $H^2\big(\mathbb{C}P^n, \mathbb{Z}\big) \cong \mathbb{Z}$. This gives that

$$H^{2k}(M, \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, \\ \mathbb{Z}/d\mathbb{Z}, & 1 \leq k \leq n, \\ 0, & \text{otherwise}, \end{cases}$$

and $\pi^*\alpha^k$ is a generator of $H^{2k}(M, \mathbb{Z})$. On the other hand,

$$c_K\big(T^{1,0}M\big) = \pi^* c_K\big(T^{1,0}\mathbb{C}P^n\big) = \left[\prod_{l=1}^m \binom{n+1}{k_l}\right] \pi^*\alpha^{|K|}.$$

Hence, for a multi-index K with $1 \leq |K| \leq n$, the K-th Chern class $c_K\big(T^{1,0}M\big)$ is equal to zero in $H^{2|K|}(M, \mathbb{Z})$ if and only if $\prod_{l=1}^m \binom{n+1}{k_l} \in d\mathbb{Z}$. In particular if we

choose d as a prime integer greater than $n+1$, then $c_K\big(T^{1,0}M\big)$ does not vanish for any K with $1 \leq |K| \leq n$. □

We next see that the degree condition in Theorem 1.1 is optimal.

Proposition 4.2. *For each positive integer n, there exists an $(n+1)$-dimensional Stein domain Ω such that its boundary $M = \partial\Omega$ satisfies $c_K\big(T^{1,0}M\big) \neq 0$ in $H^{2|K|}(M,\mathbb{C})$ for any multi-index K with $1 \leq 2|K| \leq n+1$.*

Proof. Let Ω_0 be a Stein domain in a two-dimensional complex manifold X_0 such that its boundary $M_0 = \partial\Omega_0$ satisfies $c_1\big(T^{1,0}M_0\big) \neq 0$ in $H^2(M_0,\mathbb{C})$; see [5, Theorem 6.2] for an example of such Ω_0. Take a defining function ρ of Ω_0 that is strictly plurisubharmonic near the closure of Ω_0. Without loss of generality, we may assume that ρ is an exhaustion function on X_0. Then, for sufficiently small ϵ, there exists a diffeomorphism $\chi\colon (-\epsilon,\epsilon)\times M_0 \to \rho^{-1}((-\epsilon,\epsilon))$ satisfying $\chi(0,p) = p$ and $\rho\circ\chi(t,p) = t$. The function $\psi_0 = -1/\rho$ gives a strictly plurisubharmonic exhaustion function on Ω_0.

We first show the statement for the case of $n = 2k-1$. Consider the domain

$$\Omega = \big\{(p_1,\dots,p_k) \in (\Omega_0)^k \,|\, \psi_0(p_1)+\cdots+\psi_0(p_k) < 2k/\epsilon\big\}.$$

The function $\psi(p_1,\dots,p_k) = \psi_0(p_1)+\cdots+\psi_0(p_k)$ is a strictly plurisubharmonic exhaustion function on $(\Omega_0)^k$, and $d\psi \neq 0$ on $M = \partial\Omega$. Hence Ω is a Stein domain in $(\Omega_0)^k \subset (X_0)^k$. As noted in the proof of Theorem 1.1, the cohomology class $c_K\big(T^{1,0}M\big)$ coincides with $c_K\big(T^{1,0}(X_0)^k|_M\big)$. Consider the map $\iota\colon (M_0)^k \to M\big(\subset (X_0)^k\big)$ defined by

$$\iota(p_1,\dots,p_k) = (\chi(-\epsilon/2,p_1),\dots,\chi(-\epsilon/2,p_k)).$$

Since this map is homotopic to the natural embedding $(M_0)^k \hookrightarrow (X_0)^k$,

$$\iota^* c_K\big(T^{1,0}M\big) = c_K\big(\iota^* T^{1,0}(X_0)^k\big) = c_K\big(T^{1,0}(X_0)^k|_{(M_0)^k}\big) = c_K\big(\big(T^{1,0}M_0\big)^k\big).$$

From the assumption on Ω_0, it follows that $c_K\big(\big(T^{1,0}M_0\big)^k\big) \neq 0$ in $H^{2|K|}\big((M_0)^k,\mathbb{C}\big)$ for any multi-index K with $1 \leq |K| \leq k$, and consequently $c_K\big(T^{1,0}M\big) \neq 0$ in $H^{2|K|}(M,\mathbb{C})$.

We next treat the case of $n = 2k$. Consider the domain

$$\Omega = \big\{(p_1,\dots,p_k,z) \in (\Omega_0)^k\times\mathbb{C} \,\big|\, \psi_0(p_1)+\cdots+\psi_0(p_k)+|z|^2 < 2k/\epsilon\big\}.$$

This Ω is a Stein domain in $(\Omega_0)^k\times\mathbb{C} \subset (X_0)^k\times\mathbb{C}$. Consider the map $\iota\colon (M_0)^k \to M = \partial\Omega$ given by

$$\iota(p_1,\dots,p_k) = (\chi(-\epsilon/2,p_1),\dots,\chi(-\epsilon/2,p_k),0).$$

Then we obtain

$$\iota^* c_K\big(T^{1,0}M\big) = c_K\big(\big(T^{1,0}M_0\big)^k\big) \neq 0$$

in $H^{2|K|}\big((M_0)^k,\mathbb{C}\big)$ for any K satisfying $1 \leq |K| \leq k$ in a similar way to the case of odd n. This proves the statement. □

Remark 4.3. Cao and Chang have stated that, for any Stein domain Ω, the first Chern class $c_1\big(T^{1,0}\Omega\big)$ vanishes in $H^2(\Omega,\mathbb{C})$; see Proposition 3.2(1) and the proof of Proposition 2.4 in [3]. This statement contradicts the above example since $c_1\big(T^{1,0}M\big)$ is the image of $c_1\big(T^{1,0}\Omega\big)$ by the restriction morphism. Unfortunately, their proof contains an error. They have claimed that the vanishing of $H^{1,1}(\Omega)$ would imply that of the first Chern class. However, this is not true since $H^2(\Omega,\mathbb{C})$ has no Hodge decomposition like on closed Kähler manifolds. In particular, we need to add some assumptions on the first Chern class to Main Theorems (1) and (2) in [3] at least.

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